

Statistical Models in Computational Biology

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Estimating emission and transition probabilities for profile HMM

Figure 1 shows an example of a profile HMM for a multiple alignment of five DNA sequences. Columns in the multiple alignment with less than 50% gaps (–) are considered match states (labeled by 1, 2, 3). The rest are considered insert states (labeled by dots).

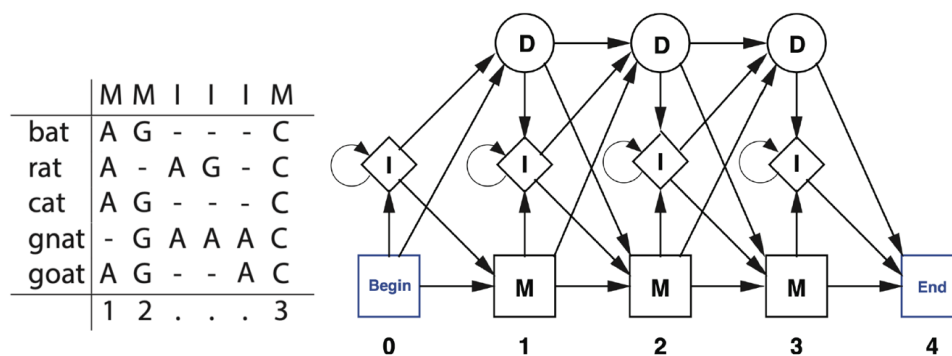


Figure 1: Example of a profile HMM (adapted from Figure 5.7 of [1])

Problem 8: Estimating emission and transition probabilities for profile HMM (3 points)

We adopt three conventions for the profile HMM on the right:

- Begin and End states are assumed as match states.
- In marked columns (M) in the alignment, gaps are assigned to delete states in the HMM (i.e., a gap is the result of a deletion).
- In unmarked columns (I) in the alignment, symbols are assigned to insert states in the HMM, and gaps are ignored.

1. We can estimate the match emission probability of symbol $a \in \mathcal{A} = \{A, C, G, T\}$ at position $i \in \{1, 2, 3\}$ by

$$E_i(a) = \frac{N_i(a) + 1}{\sum_{a' \in \mathcal{A}} (N_i(a') + 1)}$$

where $N_i(a)$ is the number of observations of the symbol a at position i across all sequences in the multiple alignment. We have added the pseudo-count 1 in order to avoid estimating some probabilities as 0.

What are the estimated match emission probabilities of the profile HMM in Figure 1?

2. Symbols appearing whilst in an insert state are considered insert emissions. Note that contiguous insert states have the same position in the model. Estimate insert emission probabilities of the profile HMM in Figure 1.

3. Let $\mathcal{S} = \{M, I, D\}$ denote the states 'Match', 'Insertion', and 'Deletion', respectively. The transition probability from state $k \in \mathcal{S}$ at position $i \in \{0, 1, 2, 3\}$ to state $\ell \in \mathcal{S}$ at position $i + 1$ can be estimated by

$$T_i(k \rightarrow \ell) = \frac{N_i(k \rightarrow \ell) + 1}{\sum_{\ell' \in \mathcal{S}} (N_i(k \rightarrow \ell') + 1)}$$

where $N_i(k \rightarrow \ell)$ is the observed number of transitions from state k at position i to state ℓ at position $i + 1$.

What are the estimated transmission probabilities in the profile HMM in Figure 1?

Problem 9: Protein family membership classification**(7 points)**

You are given multiple alignments of protein sequences for two protein families: GTP binding proteins, and a family of ATPases. The task is to parametrize profile hidden Markov models for the two protein families, and use the models to determine the family membership of unclassified protein sequences. Please use the partially provided R code in "profileHMM.R" for this question. Please include all your code and results in an R markdown file for submission.

1. Read the two alignments 'GTP_binding_proteins.txt' and 'ATPases.txt' into memory using the function `parseAlignment()` from "profileHMM.R".
2. Use the function `learnHMM()` from "profileHMM.R" to parametrise two profile HMMs: one for each protein family (multiple alignment).
3. For each family, identify the position(s) with the highest match and with the highest insert emission frequencies over all symbols. Plot the respective match and insert emission frequencies for the identified positions. (1 point)
4. The function `forward()` in "profileHMM.R" takes as input a profile HMM $\mathcal{M}$ and a sequence x . It returns the log odds ratio

$$\log \frac{P(x | \mathcal{M})}{P(x | \mathcal{R})}$$

of the probability of observing the sequence x given the model $\mathcal{M}$ versus the probability of observing the sequence x given the random model $\mathcal{R}$. Complete the `forward()` function by filling out the recursion step. (3 points)

5. The file `Unclassified_proteins.txt` contains 31 protein sequences from unknown families. Load the protein sequences into a list using the `parseProteins()` function from in "profileHMM.R".
6. For each unclassified protein $x^{(i)}$ in the list, apply the forward algorithm for both models M_1 and M_2 to obtain the log odds ratio

$$q(x^{(i)}) := \log \left(\frac{P(x^{(i)} | M_1)}{P(x^{(i)} | M_2)} \right).$$

Plot the values $q(x^i)$. Which proteins in the list belong to which family? Can you clearly decide for each protein? (3 points)

References

- [1] Durbin, R. et al. *Biological Sequence Analysis*. Cambridge University Press, 1998, <https://doi.org/10.1017/CB09780511790492>